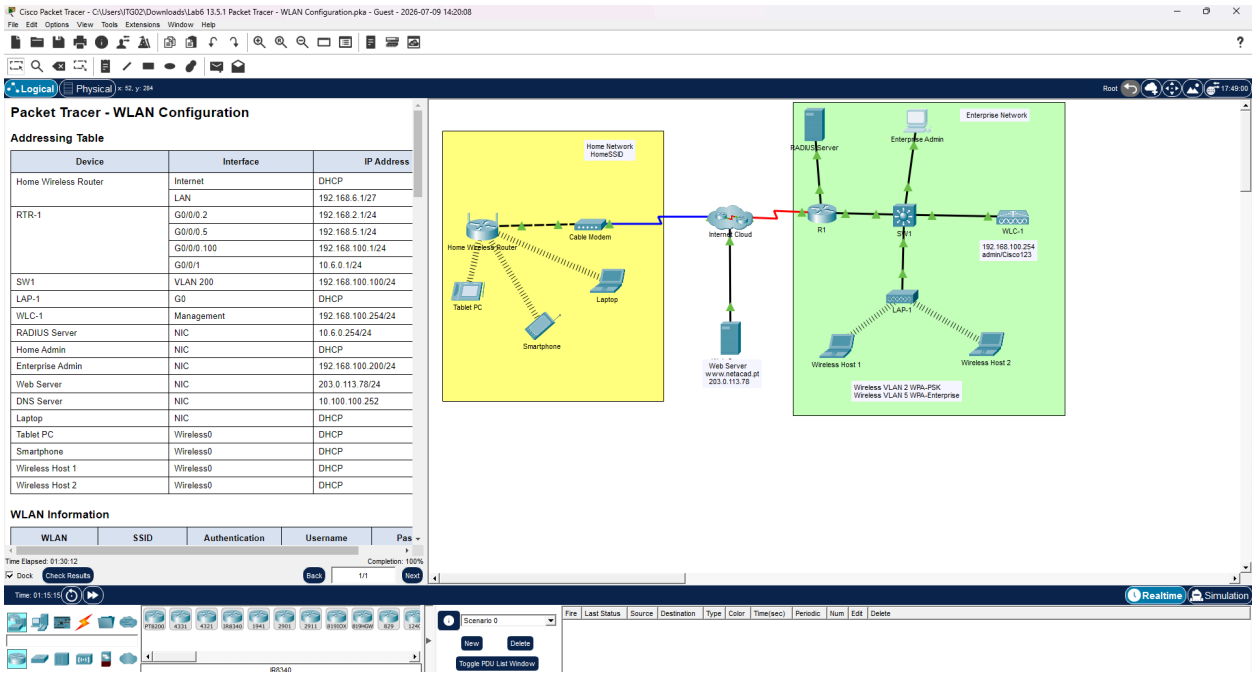
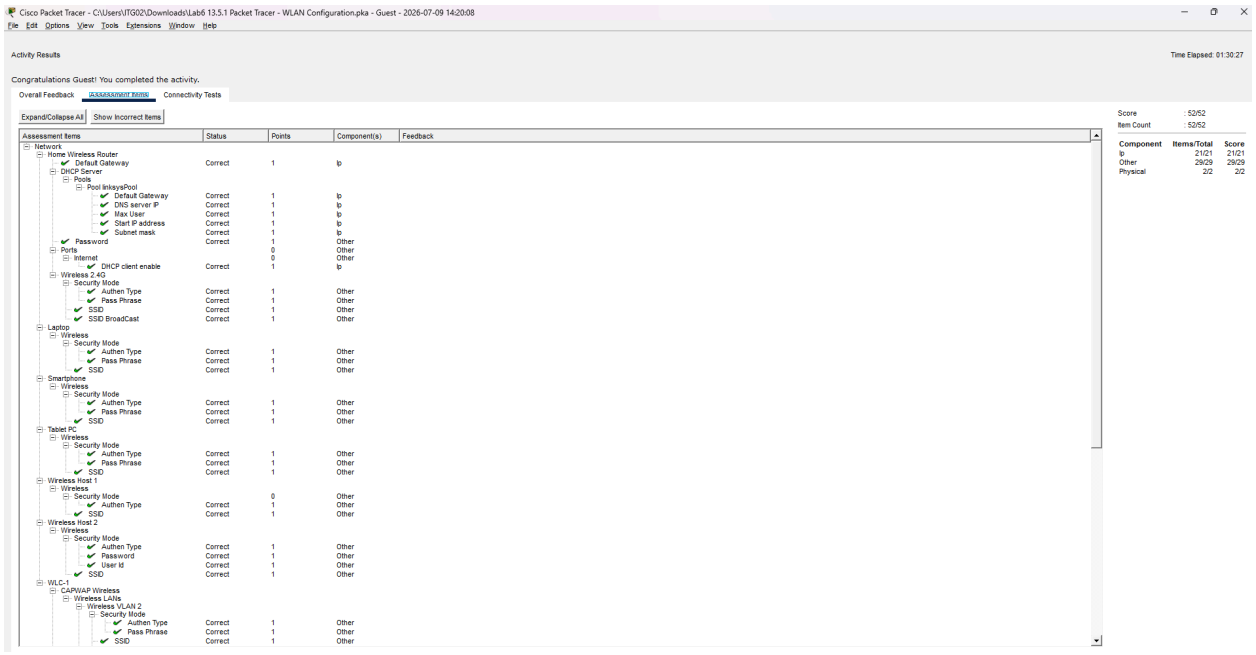
Lab Report: 13.5.1 Packet Tracer - WLAN Configuration







1. Objectives

The primary goal of this activity is to configure a hybrid wireless environment featuring both a home-grade wireless router and an enterprise-grade Wireless LAN Controller (WLC). Key tasks include:

- Configuring a home router with WPA2-PSK security and DHCP settings.
- Implementing WLC-based networking with multiple VLAN-mapped WLANs.
- Deploying WPA2-Enterprise security using a RADIUS server.
- Verifying end-to-end connectivity across various wireless client types.

2. Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Home R...	Internet	DHCP	Assigned by ISP	N/A
Home R...	LAN	192.168.6.1	255.255.255.224	N/A
RTR-1	G0/0/0.2	192.168.2.1	255.255.255.0	N/A

Device	Interface	IP Address	Subnet Mask	Default Gateway
RTR-1	G0/0/0.5	192.168.5.1	255.255.255.0	N/A
WLC-1	Management	192.168.100.254	255.255.255.0	192.168.100.1

3. WLAN Information

WLAN	SSID	Authentication	Credentials
Home Network	HomeSSID	WPA2-Pers...	Pass: Cisco123
WLAN VLAN 2	SSID-2	WPA2-Pers...	Pass: Cisco123
WLAN VLAN 5	SSID-5	WPA2-Enter...	User: userW...

4. Router & Controller Configurations

4.1 Home Wireless Router Setup

- **LAN Settings:** IP address updated to 192.168.6.1 with a /27 mask (255.255.255.224).
- **DHCP Server:** Configured to start at .3 with a maximum of 20 users.
- **Wireless:** SSID "HomeSSID" enabled on 2.4GHz, Channel 6.
- **Security:** WPA2-AES Personal enabled with passphrase "Cisco123".

4.2 Enterprise WLC-1 Configuration

- **Interfaces:** Created virtual interfaces "WLAN 2" (VLAN 2) and "WLAN 5" (VLAN 5) mapped to Port 1.
- **DHCP Management Scope:** Configured pool 192.168.100.235 – 192.168.100.245 for AP management.
- **RADIUS Server:** Added server at 10.6.0.254 with shared secret "RadiusPW".

- **SNMP:** Community string "WLAN" configured for logging to 10.6.0.254.

5. Lab Questions and Answers

Part 1, Step 1, d: Configure the internet interface of the router to receive its IP address over DHCP.

- **Question:** Verify the address. What address did it receive?
- **Answer:** In the Packet Tracer simulation, the Home Wireless Router receives its Internet (WAN) IP address from the ISP server. Typically, this address is in the **200.200.200.x/24** range (e.g., 200.200.200.226), depending on the specific instance of the lab environment provided by the instructor.

6. Connectivity Verification

Following configuration, the following results were achieved:

1. **Laptop and Tablet:** Successfully associated with "HomeSSID" and received DHCP addresses in the 192.168.6.x range.
2. **Wireless Host 1:** Successfully connected to "SSID-2" using WPA2-PSK and confirmed ping to Web Server.
3. **Wireless Host 2:** Successfully authenticated via RADIUS for "SSID-5" using Enterprise credentials.
4. **Web Access:** All clients confirmed successful HTTP/HTTPS access to the Cisco Web Server URL.